CALCULO INTEGRAL BMA02
semana 5

Método de integración:
De irracionales binomiales

$$I = \int x^m[a + bx^n]^p dx$$

Método de Chebyshev

$m, n, y\ p\ son\ racionales$

$$I = \int x^m[a + bx^n]^p dx$$

$$caso\ 1\!: \frac{m+1}{n} \in \mathbb{Z} \Rightarrow x^n = t \Rightarrow nx^{n-1}dx = dt$$

$$I = \frac{1}{n}\int t^{\frac{m+1}{n}-1}[a + bt]^p dt \Rightarrow a + bt = z^r \qquad r\ es\ el\ denominador\ de\ p$$

$$Ejercicio\ 1)\ calcule\!: \ I = \int x^{-7}[65 - x^6]^{-\frac{1}{6}}dx \Rightarrow m = -7, n = 6, p = -\frac{1}{6}$$

$$\frac{m+1}{n} = -1 \in \mathbb{Z} \Rightarrow 65 - x^6 = z^6 \Rightarrow -6x^5dx = 6z^5dz \Rightarrow I = \int x^{-7}[z^6]^{-\frac{1}{6}}\left(-\frac{z^5}{x^5}\right)dz$$

$$I = -\int x^{-12}z^4dz = -\int [65 - z^6]^2 z^4 dz = -\int (4225z^4 - 130z^{10} + z^{16})dz$$

$$I = \frac{130}{11}\sqrt[6]{(65 - x^6)^{11}} - 845\sqrt[6]{(65 - x^6)^5} - \frac{1}{17}\sqrt[6]{(65 - x^6)^{17}} + C$$

Método de Chebyshev

$m, n, y\ p\ son\ racionales$

$$I = \int x^m[a + bx^n]^p dx$$

$caso\ 2: (\frac{m+1}{n} + p) \in \mathbb{Z}$ ⇒ $x^n = t$ ⇒ $nx^{n-1}dx = dt$

$I = \frac{1}{n}\int t^{\frac{m+1}{n}-1}[a + bt]^p dt$ ⇒ $at^{-n} + b = z^r$ $r\ es\ el\ denominador\ de\ p$

$Ejercicio\ 2)\ calcule{:}\ I = \int t^2[1 + t]^{\frac{1}{2}}dt$ ⇒ $m = 2, n = 1, p = \frac{1}{2}$

$1 + t = u^2$ ⇒ $dt = 2udu$ ⇒ $I = \int [u^2 - 1]^2[u^2]^{\frac{1}{2}}2udu = 2\int (u^6 - 2u^4 + u^2)du$

$$I = 2\left[\frac{u^7}{7} - \frac{2u^5}{5} + \frac{u^3}{3}\right] + C = \left[\frac{2\left(\sqrt{1+t}\right)^7}{7} - \frac{4\left(\sqrt{1+t}\right)^5}{5} + \frac{2\left(\sqrt{1+t}\right)^3}{3}\right] + C$$

Método de Chebyshev

$$Ejercicio\ 3)\ calcule:\ I=\int\frac{\sqrt[3]{1+\sqrt[4]{x}}}{\sqrt{x}}dx$$

$$I=\int x^{-\frac{1}{2}}\left[1+x^{\frac{1}{4}}\right]^{\frac{1}{3}}dx \Rightarrow m=-\frac{1}{2}2,n=\frac{1}{4}1,p=\frac{1}{3} \Rightarrow \frac{m+1}{n}=2$$

$$1+x^{\frac{1}{4}}=u^3 \Rightarrow \frac{1}{4}x^{-\frac{3}{4}}dx=3u^2du \qquad I=\int x^{-\frac{1}{2}}[u^3]^{\frac{1}{3}}12u^2x^{\frac{3}{4}}du$$

$$I=12\int x^{\frac{1}{4}}u^3du=12\int[u^3-1]\,u^3du=12\left(\frac{u^7}{7}-\frac{u^4}{4}\right)+C$$

$$I=12\left(\frac{\left[\sqrt[3]{1+\sqrt[4]{x}}\right]^7}{7}-\frac{\left[\sqrt[3]{1+\sqrt[4]{x}}\right]^4}{4}\right)+C$$

Método de Chebyshev

$$I = \int x^m [a + bx^n]^{\frac{p}{q}} dx$$

$m, n, y\ p\ son\ racionales$

$$caso\ 2: (\frac{m+1}{n} + \frac{p}{q}) \in \mathbb{Z} \Rightarrow u = \left[\frac{a + bx^n}{x^n}\right]^{\frac{1}{q}} \Rightarrow u^q = \left[\frac{a + bx^n}{x^n}\right]$$

$$Ejercicio\ 4)\ calcule:\ I = \int x^2 [3 + x^4]^{\frac{1}{4}} dx \Rightarrow u^4 = \left[\frac{3 + x^4}{x^4}\right] \Rightarrow 4u^3 du = [-12x^{-5}]dx$$

$$I = \int x^2 [u^4 x^4]^{\frac{1}{4}} (-\frac{4}{12} x^5) u^3 du = -\frac{1}{3} \int x^8 u^4\, du \qquad u^4 = 1 + \frac{3}{x^4} \qquad u^4 - 1 = \frac{3}{x^4}$$

$$I = -\frac{1}{3} \int \left[\frac{3}{u^4 - 1}\right]^2 u^4\, du \Rightarrow I = -3 \int \left[\frac{1}{u^4 - 1}\right]^2 u^4\, du \Rightarrow$$

FIEE

PROBLEMA 1

$$I_n = \int \frac{\mathrm{d}x}{(1+x^2)^n}$$

Sea: $x = \tan\theta \quad \theta = \arctan x \to \mathrm{d}\theta = \dfrac{\mathrm{d}x}{1+x^2}$

$$dx = \sec^2\theta \, d\theta$$

Reemplazando en la Integral

$$I_n = \int \frac{\sec^2\theta \, d\theta}{(\sec^2\theta)^n}$$

$$u = \sec^{-2n}\theta \to du = (-2n)\sec^{-2n}\theta(\tan\theta)\, d\theta$$

$$\mathrm{d}v = \sec^2\theta \, \mathrm{d}\theta \to v = \tan\theta$$

Integración por partes

$$I_n = \sec^{-2n}\theta \tan\theta - \int \tan\theta(-2n\sec^{-2n}\theta\tan\theta\,\mathrm{d}\theta)$$

$$I_n = \sec^{-2n}\theta\tan\theta + 2\mathrm{n}\int \sec^{-2n}\theta\tan^2\theta\,\mathrm{d}\theta$$

$$I_n = \sec^{-2n}\theta\tan\theta + 2n\int \sec^{-2n}\theta(\sec^2\theta - 1)\, d\theta$$

$$I_n = \sec^{-2n}\theta\tan\theta + 2n(\int \sec^{-2n}\theta\sec^2\theta\, d\theta - \int \sec^{-2n}\theta\, d\theta)$$

$$I_n = \sec^{-2n}\theta\tan\theta + 2nI_n - 2n\int\cos^{2n}\theta\, d\theta$$

$$I_n = \frac{1}{1-2n}\cdot\frac{x}{(1+x^2)^n} - \frac{2n}{1-2n}I_{n+1}$$

$$Si\ x = tan\theta\ ,\cos\theta = \frac{1}{\sqrt{x^2+1}}$$

Reemplazando la integral en funcion de x

$$I_n = \frac{x}{(1+x^2)^n} + 2nI_n - 2n\frac{dx}{(1+x^2)^n(1+x^2)}$$

$$I_n = \frac{x}{(1+x^2)^n} + 2nI_n - 2nI_{n+1}$$

PROBLEMA 2

$$I = \int \frac{dx}{(x^2 - 4x + 8)^{3/2}}$$

1er Método

$$I = \int \frac{dx}{((x-2)^2 + 4)^{3/2}}$$

Sea:

$$x - 2 = 2\tan\theta$$
$$dx = 2\sec\theta^2 d\theta$$

$$I = \int \frac{2\sec\theta^2 d\theta}{(4\sec\theta^2)^{3/2}}$$

$$I = \frac{2}{8}\int \frac{\sec\theta^2 d\theta}{sec\theta^3}$$

$$I = \frac{1}{4}\int Cos\theta\, d\theta$$

$$I = \frac{1}{4} Sen\theta + C$$

$$\theta = \arctan(\frac{x-2}{2})$$

$$\boldsymbol{I = \frac{1}{4} Sen(\arctan\left(\frac{x-2}{2}\right)) + C}$$

$$I = \int \frac{dx}{(x^2 - 4x + 8)^{3/2}}$$

2do Método

$$I = \int \frac{dx}{((x-2)^2 + 4)^{3/2}}$$

Sea: $x - 2 = 2senh\theta$

$dx = 2cosh\,\theta d\theta$

Recordar que: $\mathrm{Cosh}^2\,\theta - \mathrm{Senh}^2\,\theta = 1$

$$I = \int \frac{2cosh\,\theta d\theta}{(4cosh\theta^2)^{3/2}}$$

$$I = \frac{2}{8}\int sech^2\theta d\theta$$

$$I = \frac{1}{4}\,\tanh\theta + \mathrm{C}$$

$$\theta = \mathrm{arcsenh}x(\frac{x-2}{2})$$

$$\boldsymbol{I = \frac{1}{4}\,\tanh(arcsenh\,x\left(\frac{x-2}{2}\right)) + C}$$

PROBLEMA 3

$$I=\int \frac{dx}{x^5\sqrt{x^2-1}}$$

$$x=\frac{1}{t}, \mathrm{x}=t^{-1}$$
$$dx=-1t^{-2}dt$$

$$I=\int \frac{-1t^{-2}dt}{t^{-5}\sqrt{t^{-2}-1}}$$

$$I=-\int \frac{t^4dt}{\sqrt{1-t^2}}=-\left((At^3+Bt^2+Ct+D)\sqrt{1-t^2}+\int \frac{\lambda}{\sqrt{1-t^2}}\right)$$

$$A=-\frac{1}{4}$$
$$B=0$$
$$\mathrm{C}=-\frac{3}{8}$$
$$D=0$$
$$\lambda=\frac{3}{8}$$

$$I=-\int\frac{t^4dt}{\sqrt{1-t^2}}=-\left(\left(\frac{-1}{4}t^3+\frac{-3}{8}t\right)\sqrt{1-t^2}+\int\frac{\frac{3}{8}}{\sqrt{1-t^2}}\right)$$

$$\int\frac{\frac{3}{8}}{\sqrt{1-t^2}}=(3/8)\big(arcsen(t)\big)$$

$$I=-\sqrt{1-t^2}\left(\frac{-1}{4}t^3+\frac{-3}{8}t\right)-(3/8)\big(arcsen(t)\big)$$

$$t=x^{-1}$$

$$\boldsymbol{I=\sqrt{1-\frac{1}{x^2}}\left(\frac{1}{4x^3}+\frac{3}{8x}\right)-\frac{3}{8}arcsen\left(\frac{1}{x}\right)+c}$$

PROBLEMA 5

$$I=\int\left(\frac{x+1}{x-1}\right)^{1/3}\frac{1}{x+1}dx=\int(x+1)^{-2/3}(x-1)^{-1/3}dx$$

$$x+1=u, du=dx$$

$$I=\int(u)^{-2/3}(u-2)^{-1/3}du$$

$$u-2=z^3u$$
$$du=2(1-z^3)^{-2}(3z^2)dz$$

$$I=\int(u)^{-2/3}(u-2)^{-1/3}du=\int 3z(1-z^3)^{-1}dz$$

$$I=3\int\frac{zdz}{1-z^3}=3\int\frac{zdz}{(1-z)(z^2+z+1)}$$

$$I=3\int\frac{1/3}{(1-z)}+3\int\frac{\left(\frac{1}{3}\right)z-1/3}{(z^2+z+1)}$$

FIEE

$$I_1 = 3\int \frac{(1/3)dz}{(1-z)} = \int \frac{dz}{1-z}$$

$$I_1 = -\ln|1-z|$$

$$I_2 = 3\int \frac{\left(\frac{1}{3}\right)z - 1/3}{(z^2+z+1)} = \frac{1}{2}\int \frac{2z-2+1-1}{(z^2+z+1)}$$

$$I_2 = \frac{1}{2}\int \frac{2z+1}{(z^2+z+1)} + \frac{1}{2}\int \frac{-3}{(z^2+z+1)}$$

$$I_2 = \frac{1}{2}\ln|(z^2+z+1)| - \frac{-3}{2}\int \frac{dz}{\left((z+1/2)^2+3/4\right)}$$

$$I_2 = \frac{1}{2}\ln|(z^2+z+1)| - \sqrt{3}\arctan\left(\frac{2z+1}{\sqrt{3}}\right)$$

$$I = I_1 + I_2 = \frac{1}{2}\ln|(z^2+z+1)| - \sqrt{3}\arctan\left(\frac{2z+1}{\sqrt{3}}\right) - \ln|1-z| \qquad z = \sqrt[3]{\frac{x-1}{x+1}}$$

$$\boldsymbol{I = \frac{1}{2}\ln\left|\left(\sqrt[3]{\frac{x-1}{x+1}}^{\,2} + \sqrt[3]{\frac{x-1}{x+1}} + 1\right)\right| - \sqrt{3}\arctan\left(\frac{2\sqrt[3]{\frac{x-1}{x+1}}+1}{\sqrt{3}}\right) - \ln\left|1 - \sqrt[3]{\frac{x-1}{x+1}}\right| + c}$$

PROBLEMA 6

$$I = \int \sqrt{x^2 + x + 2 + \sqrt{2x^3 + 7x^2 + 4x + 3}}\,\mathrm{d}x$$

$\underline{Aplicando\ la\ formula\ de\ radicales\ dobles}$:

$$\sqrt{A \pm \sqrt{B}} = \sqrt{\frac{A + C}{2}} \pm \sqrt{\frac{A - \mathrm{C}}{2}}$$

$$Donde: C = \sqrt{A^2 - B}$$

$$Sea: A = x^2 + x + 2 \quad \text{y} \quad \mathrm{B} = 2x^3 + 7x^2 + 4x + 3$$

$$C = \sqrt{(x^2 + x + 2)^2 - (2x^3 + 7x^2 + 4x + 3)}$$

$$C = x^2 - 1$$

$$I = \int \left(\sqrt{\frac{x^2 + x + 2 + x^2 - 1}{2}} + \sqrt{\frac{x^2 + x + 2 - x^2 + 1}{2}} \right) \mathrm{dx}$$

$$I = \int \frac{1}{\sqrt{2}} \left(\sqrt{2x^2 + x + 1} + \sqrt{x + 3} \right) dx$$

$$Sea:\ I_1 = \int \sqrt{2x^2 + x + 1}\,\mathrm{d}x \quad y \quad I_2 = \int \sqrt{x + 3}\,dx$$

$$I_1 = \int \sqrt{2x^2 + x - 1}\, \mathrm{d}x$$

$$I_1 = \sqrt{2} \int \sqrt{\left(x + \frac{1}{4}\right)^2 + \frac{7}{16}}\, dx$$

$$Sea: x + \frac{1}{4} = \frac{\sqrt{7}}{4} \tan\theta \rightarrow \mathrm{d}x = \frac{\sqrt{7}}{4} \sec^2\theta\, \mathrm{d}\theta$$

Reemplazando y operando:

$$I_1 = \frac{7\sqrt{2}}{16} \int \sec^3\theta$$

$$I_1 = \frac{7\sqrt{2}}{32} (\sec\theta \cdot \tan\theta + \ln|\tan\theta + \sec\theta|) + C$$

$$I_1 = \frac{7\sqrt{2}}{32} \left(\frac{\sqrt{(4x+1)^2 + 7} \cdot (4x+1)}{7} + \ln\left| \frac{\sqrt{(4x+1)^2 + 7} + 4x + 1}{\sqrt{7}} \right| \right) + C$$

$$I_2 = \int \sqrt{x + 3}\, \mathrm{d}x$$

$$Sea: x + 3 = u \rightarrow \mathrm{d}x = \mathrm{d}u$$

$$I_2 = \int u^{1/2}\, \mathrm{d}u = \frac{2}{3} u^{3/2} + \mathrm{C}$$

$$I_2 = \frac{2}{3}(x+3)^{3/2} + C$$

$$Entonces: I = \frac{1}{\sqrt{2}}(I_1 + I_2)$$

$$I = \frac{1}{\sqrt{2}}\left(\frac{7\sqrt{2}}{32}\left(\frac{\sqrt{(4x+1)^2+7}\cdot(4x+1)}{7} + \ln\left|\frac{\sqrt{(4x+1)^2+7}+4x+1}{\sqrt{7}}\right|\right) + \frac{2}{3}(x+3)^{3/2}\right) + C$$